QuantRule - Architecture Characteristics

QuantRule

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Eight characteristics matter. Each is tied to a numbered requirement and has a level that can be measured. The rest are deliberately not optimised.

Characteristic	Kata	Why it matters	Level (measurable)	Measured by
Testability	R2, R3, R5	every published number is a claim someone defends	every reading and event has a causality, a determinism, and a golden test; fast suite <= 5 min	`cargo test`, gate
Modularity	R4, R6, R7	six services only pay off if boundaries hold	6 services, 0 cycles, gateway imports 0 engine symbols	boundary test
Maintainability	R3, R7	the rule changes weekly, the platform must not	a rule change touches <= 1 crate and <= 3 modules; no file over 1,000 lines	structural check
Deployability	R4, R6	a reviewer must reproduce in one command	clean clone to healthy <= 10 min; 1 published port; desktop is 1 asset	`make up`
Fault tolerance	R4, R5	a run must survive a crash without corrupting evidence	0 succeeded runs with a missing artifact; resubmission gives identical bytes	crash test
Responsiveness	R4, R7	marking is interactive, a year of candles is not	submit <= 200 ms; chart window <= 200 ms; 1 year, 8 timeframes <= 60 s on 8 cores	benchmark
Scalability	R4	one machine must borrow a second	model stage 1 to 4 workers: identical digest, >= 2.5x throughput	scale test
Partitioning type	R1-R7	capability boundaries keep a rule change inside one service	domain-driven: 6 services by capability, no layer split	service catalog

Not optimised: elasticity (no burst traffic, one operator, batch work), localization (one language, one user), and accessibility (one expert user; keyboard support is kept, but it is not a driver).